

Def. Let  $G$  be a group, and  $H \leq G$  be a subgroup,  
 Given an element  $g \in G$ , define left-Coset of  $H$  as  

$$gH \stackrel{\text{def}}{=} \{gh \mid h \in H\}.$$

Lemma. Let  $G$  be a group, and  $H \leq G$  be a subgroup. Then, for each  $g_1, g_2 \in G$ ,  
 $g_1H = g_2H$  if and only if  $g_1^{-1}g_2 \in H$ .

Proof. Given  $g_1, g_2 \in G$ , suppose that  $g_1H = g_2H$ . Then, since  $g_2 \in g_2H = g_1H$  ( $1 \in H$ ),  
 $g_2 = g_1h$  for some  $h \in H$ , thus  $g_1^{-1}g_2 \in H$ .

Conversely, let  $g_1, g_2 \in G$  be given st  $g_1^{-1}g_2 \in H$ . That is,  $g_1h = g_2$  for some  $h \in H$ .  
 Given  $g_1h_1 \in g_1H$ ,  $g_1h_1 = g_2h^{-1}h_1 \in g_2H$ . Given  $g_2h_2 \in g_2H$ ,  $g_2h_2 = g_1hh_2 \in g_1H$   $\square$

Q. In left Coset sense,  $g_1H = g_2H \Leftrightarrow g_1g_2^{-1} \in H$  hold?

A. No!, Ss,  $H = \langle (1, 2) \rangle$ ,  $g_1 = (1\ 3)$  and  $g_2 = (1\ 2\ 3)$

Thm. Lagrange Thm

If  $H$  is a subgroup of a group  $G$ , then  $|G|/|H| = [G:H]$

Proof. Given a subgroup  $H \leq G$ , define a relation on  $G$  as;

$$x \sim y \Leftrightarrow xH = yH$$

Then, it is an equivalence relation, so a Collection  $\{gH \mid g \in G\}$  forms a partition.  
 Moreover, for each  $g \in G$ , a map

$$h \mapsto gh \quad (h \in H)$$

is onto by definition and one to one by the cancellation law, so  $|H| = |gH|$ .

Isomorphism Theorem.

1st Isomorphism theorem.

For any Homomorphism  $\varphi: G \rightarrow G'$ ,

$$G/\ker \varphi \cong \operatorname{Im} \varphi.$$

+ If  $G$  is finite, then  $|G| = |\ker \varphi| \cdot |\operatorname{Im} \varphi|$ .

Proof. Since  $\ker \varphi \trianglelefteq G$ ,  $G/\ker \varphi$  is well-defined as a group. Define

$$\tilde{\varphi}: G/\ker \varphi \rightarrow \operatorname{Im} \varphi: g\ker \varphi \mapsto \varphi(g).$$

Then,  $g_1\ker \varphi = g_2\ker \varphi \Leftrightarrow g_1g_2^{-1} \in \ker \varphi \Leftrightarrow \varphi(g_1) = \varphi(g_2)$

so  $\tilde{\varphi}$  is well-defined and injective, and the surjective is clear. ~~is~~

Note: the First Isomorphism theorem gives:

$N \trianglelefteq G$  is Normal subgroup if and only if  $\pi: G \rightarrow G/N$  is Homomorphism,  $\ker \pi = N$ .

Prop. Let  $\varphi: G \rightarrow G'$  be a Homomorphism.

- 1).  $H \leq G \Rightarrow \varphi[H] \leq G'$
- 2).  $N \trianglelefteq G \Rightarrow \varphi[N] \trianglelefteq \varphi[G] = \operatorname{Im} \varphi$ .
- 3).  $H' \leq G' \Rightarrow \varphi^{-1}[H'] \leq G$
- 4).  $N' \trianglelefteq G' \Rightarrow \varphi^{-1}[N'] \trianglelefteq G$ .

2<sup>nd</sup> Isomorphism theorem.

Let  $G$  be a group, and let  $A, B$  be subgroups of  $G$  satisfying  $A \leq N_G(B)$ . Then,  $AB \leq G$  and  $B \trianglelefteq AB$ ,  $A \cap B \trianglelefteq A$ . Further,

$$AB/B \cong A/A \cap B$$

Proof. Since  $A$  normalizes  $B$ ,

$$AB = \bigcup_{a \in A} aB = \bigcup_{a \in A} Ba = BA$$

and so given  $a_1b_1, a_2b_2 \in AB$ ,  $a_1b_1(a_2b_2)^{-1} = a_1b_1b_2^{-1}a_2^{-1} \in AB$ , so  $AB \leq G$ .  $B \leq AB$  is trivial and given  $ab \in AB$  and  $g \in B$ ,  $abgb^{-1}a^{-1} \in B$ , so  $B \trianglelefteq AB$ .  $A \cap B \trianglelefteq A$  is also trivial (These are given by the fact that  $A$  normalizes  $B$ ).

Now, define a map

$$Q: A \rightarrow AB/B: a \mapsto aB.$$

It is well-defined since  $A \leq AB$  and it is Homomorphism because: given  $a_1, a_2 \in A$ ,

$$Q(a_1a_2) = a_1a_2B = a_1B \cdot a_2B$$

and onto: given  $abB \in AB/B$ , take  $a \in A$ , then  $Q(a) = aB = abB$ .

And, the kernel is  $\ker Q = \{g \in A: gB = B\} = \{g \in A: g \in B\} = A \cap B$ . Consequently, the first isomorphism theorem gives  $A/A \cap B \cong AB/B$ .  $\square$

### 3rd Isomorphism Theorem.

Let  $G$  be a group and  $H, K \trianglelefteq G$  be normal subgroups with  $H \leq K$ .  
Then,  $K/H \trianglelefteq G/H$  and  $[G/H]/[K/H] \cong G/K$ .

Proof. Since  $K \trianglelefteq G$ ,  $K/H \trianglelefteq G/H$  directly. Define a map  
 $\varphi: G/H \rightarrow G/K: gH \mapsto gK$ .

It is well-defined, because: given  $g_1, g_2 \in G$ ,

$$g_1H = g_2H \Leftrightarrow g_1g_2^{-1} \in H \Rightarrow g_1g_2^{-1} \in K \Leftrightarrow g_1K = g_2K = \varphi(g_1) = \varphi(g_2)$$

$\uparrow$   
 $H \leq K$  (so, injectiveness is given only  $H=K$ ).

Surjectivity and Homomorphism are clear. Moreover,

$$\ker \varphi = \{gH \in G/H \mid gK = K\} = \{gH \in G/H \mid g \in K\} = K/H.$$

Finally, the 1st isomorphism theorem gives the result.

14th Isomorphism Theorem.

Homomorphism.

Def. Let  $G, H$  be groups. A map  $\varphi: G \rightarrow H$  is called a homomorphism if for all  $g_1, g_2 \in G$ ,  $\varphi(g_1 \cdot_G g_2) = \varphi(g_1) \cdot_H \varphi(g_2)$ .

Basic Properties.

$\varphi(g^n) = \varphi(g)^n$  for any  $g \in G$  and  $n \in \mathbb{Z}$ . So,  $|\varphi(g)| \mid |g|$ .